

Week / Sprint	Sprint Worklets / Tasks	Applicable Core Subject & Concepts	KPI's
Week 1	• Elicit project requirements • Draft SRS • Prepare use-case model • Design registration UI • Establish DB connectivity • Prepare registration test cases • Conduct team review	OOP: • Classes & Objects • Abstraction DBMS: • ER Model • Entities & Relationships • Database Connectivity Java: • Classes & Methods • JDBC basics Software Engineering: • SRS • Use Cases • Requirements Analysis • Testing Fundamentals	• 100% registration requirements documented • SRS + use-case diagram completed • 2 registration screens completed • DB connection demonstrated • ≥15 test cases prepared • ≥90% planned tasks completed
Week 2	• Design user-table ER/database • Develop login UI • Implement client-side validation • Develop authentication logic • Implement password hashing • Perform boundary/negative testing	DBMS: • ER Diagram • Primary/Foreign Keys • Constraints• SQL OOP: • Encapsulation • Classes & Objects Java: • Exception Handling • Methods Cybersecurity: • Authentication • Password Hashing • Access Control Software Testing: • Boundary Testing • Negative Testing	• ER diagram and user schema approved • Login UI completed • Valid-user authentication successful • ≥20 login test cases executed • 100% critical authentication defects logged • ≥90% test cases pass

Week 3	• Design profile UI/UX • Implement profile view/edit workflow • Develop update queries • Implement validation • Integrate registration → login → profile • Perform regression testing	DBMS: • CRUD Operations • UPDATE Queries • Constraints OOP: • Encapsulation • Methods • Object Interaction Java: • Collections • Exception Handling Software Testing: • Integration Testing • Regression Testing • End-to-End Testing	• Profile module completed • CRUD/update flow demonstrated • Registration → login → profile flow passes • ≥95% integration test pass rate • 0 open critical defects
Week 4	• Gather alumni-search requirements • Prepare search-page wireframe/UI • Retrieve alumni data • Implement Linear Search • Implement Binary Search • Validate keywords • Conduct accuracy/performance testing	Data Structures: • Arrays • Searching • Linear Search • Binary Search • Time Complexity DBMS: • SELECT Queries • WHERE Clause • Indexing basics OOP: • Methods • Modular Design Java/C/C++: • Search Algorithm Implementation	• Search UI completed • ≥95% search accuracy • Search response meets agreed target • ≥15 search test cases executed • 100% required search fields validated
Week 5	• Design mentor-request schema • Develop send-request UI • Implement request validation • Implement DB insert/update logic • Handle duplicate requests	DBMS: • Table Design • INSERT/UPDATE • Primary Keys • Unique Constraints OOP: • Classes & Objects • Encapsulation Data Structures: • Lists • Sets	• Request table and UI completed • ≥95% valid requests stored successfully • Duplicate/invalid requests handled • ≥20 unit/negative tests executed • ≥90% pass rate